Malaria Cell Classification System
MLOps Project Report

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April 28, 2026

1 Introduction

This project was developed as a complete AI application for malaria cell image classification. The repository was organized to reflect the full lifecycle of a real ML system: data preparation, experiment tracking, model serving, monitoring, feedback collection, retraining, and deployment packaging.

The application supports both technical and non-technical use. A clinician or evaluator can upload cell images through a web interface and receive predictions, while the engineering side of the system keeps track of model versions, runtime health, drift signals, feedback queues, and retraining activity.

2 Project Objective and Scope

The main goal of the project is to classify malaria cell images into two categories: *Parasitized* and *Uninfected*. To make that goal practical and sustainable, the repository was built around a broader set of objectives. The system needed to provide a frontend for inference, backend APIs for prediction and operations, a reproducible DVC workflow for training and evaluation, MLflow-based tracking and model versioning, Airflow-based retraining for newly labelled data, and live monitoring through Prometheus and Grafana.

3 Dataset Overview

The project uses the Cell Images for Detecting Malaria dataset, which is a publicly available collection of microscopic images of red blood cells. These images were captured using thin blood smear slides and are commonly used in malaria diagnosis workflows. The dataset consists of two balanced classes: Parasitized: Images containing red blood cells infected with malaria parasites. Uninfected: Images of healthy red blood cells without infection. In total, the dataset contains 27,558 images, evenly split between the two categories, making it suitable for supervised classification tasks without severe class imbalance issues.

Each image is: A colored microscopic image Typically resized to a fixed resolution (commonly 64×64 or 128×128 in preprocessing pipelines) Stored in a directory structure where folder names correspond to class labels

link: https://www.kaggle.com/datasets/iarunava/cell-images-for-detecting-malaria

4 Repository Organisation

The key top-level components are:

- `src/`: backend logic, model training, evaluation, feedback processing, MLflow utilities, and serving logic.
- `frontend/`: Streamlit user interface.
- `airflow/`: Airflow DAGs and containerization for orchestration.
- `docker/`: Dockerfiles and service entrypoints.
- `docs/`: architecture, high-level design, low-level design, test plan, user manual, and reproducibility notes.
- `monitoring/`: Grafana provisioning and Alertmanager configuration.
- `dvc.yaml` and `dvc.lock`: reproducible ML pipeline definition.
- `docker-compose.yaml`: multi-service deployment definition.

4.1 Technology Stack

The current repository uses the following tools and frameworks:

- Frontend: Streamlit
- Backend APIs: FastAPI
- Deep Learning: PyTorch and Torchvision
- Experiment Tracking and Registry: MLflow
- Reproducible Pipeline: DVC
- Workflow Orchestration: Apache Airflow
- Monitoring: Prometheus
- Visualization: Grafana
- Alert Routing: Alertmanager
- Containerization: Docker and Docker Compose

5 System Architecture

The system follows an architecture in which the frontend and backend are separate software blocks connected through REST APIs.

5.1 High-Level Design

Major Components

1. **Frontend**: Built with Streamlit, used by non-technical users to upload cell images and view predictions, alerts, live metrics, pipeline state, and retraining controls.
2. **Backend**: Built with FastAPI, exposes inference, monitoring, failure-log, pipeline-status, feedback-statistics, and retraining APIs. Loads the current `Production` model from MLflow and stores uncertain/OOD samples into feedback queues.
3. **Model Registry and Experiment Tracking**: MLflow tracks runs, metrics, artifacts, and model versions. Production inference always points to the `MalariaClassifier` model in the `Production` stage.
4. **Reproducible Training Pipeline**: DVC defines the full pipeline (data preparation, EDA, preprocessing, resizing, training, evaluation). `dvc repro` provides reproducible execution.
5. **Feedback Retraining Pipeline**: Airflow orchestrates incremental retraining using newly labelled feedback data. The DAG periodically checks for new samples, processes them, fine-tunes the model, evaluates performance, and promotes the updated model.
6. **Monitoring and Alerting**: Prometheus collects metrics, Grafana visualizes them, and Alertmanager forwards alerts back to the backend for UI display.

Design Rationale

- **Loose coupling**: Frontend and backend interact only through REST APIs.
- **Environment parity**: Deployment is standardized using Docker and Docker Compose.
- **Operational transparency**: Monitoring, alerting, and retraining states are visible in the UI.
- **Human-in-the-loop**: OOD and uncertain samples are reviewed by domain experts before retraining.
- **Reproducibility**: Pipelines and experiments are tracked using DVC and MLflow.

Logging

The backend logs:

- prediction flow
- evaluation metrics
- retraining events

Logging supports debugging, monitoring, and traceability.

5.2 Low-Level Design

Primary API Endpoints

Endpoint	Method	Purpose	Input	Output
`/health`	GET	Liveness check	None	{status: ok}
`/ready`	GET	Readiness check	None	{ready: true}
`/predict`	POST	Run inference	Image file	Prediction + confidence + flags
`/api/metrics`	GET	Fetch live metrics	None	Aggregated stats
`/api/failures`	GET	View failure logs	Optional filters	Failure entries
`/api/feedback/stats`	GET	Feedback queue stats	None	Counts of samples
`/pipeline/status`	GET	DVC status	None	Status output
`/retrain`	POST	Trigger DVC pipeline	None	Execution logs
`/api/retraining/status`	GET	Airflow status	None	DAG run info
`/api/retraining/trigger`	POST	Trigger DAG	Optional config	Run ID + state
`/model/info`	GET	Model metadata	None	Version + metrics
`/alerts`	GET	Active alerts	None	Alert list
`/internal/alert`	POST	Alert webhook	Payload	Acknowledgement

Inference Flow

```
Upload image → Validate → Preprocess → Run model →
Compute metrics → Log → Save uncertain/OOD → Return response
```

DVC Pipeline Stages

1. prepare_raw_data
2. eda
3. preprocess
4. resize
5. train

6. evaluate

Airflow Feedback DAG Tasks

1. check_new_images
2. process_feedback
3. finetune_model
4. evaluate_finetune
5. promote_model
6. dvc_snapshot

Error Handling

- 400: invalid or unreadable image
- 422: unsupported file type
- 503: model not loaded
- 500: internal server error

Example Request/Response

```
POST /predict

Response:
{
  "prediction_label": "Parasitized",
  "confidence": 0.91,
  "entropy": 0.32,
  "is_ood": false,
  "low_confidence": false
}
```

5.3 Containerized Deployment Architecture

The deployment is defined through `docker-compose.yaml`. The Compose stack includes the following important services:

- `frontend`
- `backend`

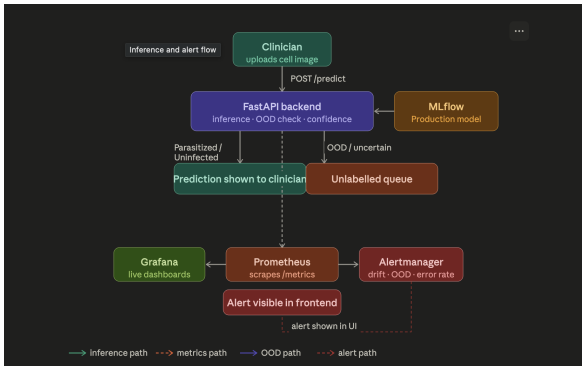
- `mlflow`
- `prometheus`
- `grafana`
- `alertmanager`
- `postgres`
- `redis`
- `airflow-apiserver`
- `airflow-scheduler`
- `airflow-dag-processor`
- `airflow-worker`
- `airflow-triggerer`

Docker Compose service stack Show terminal output of `docker compose ps` with frontend, backend, mlflow, prometheus, grafana, alertmanager, and Airflow services running. fig-docker-compose-services.png

6 System Design

6.1 Overall Architecture

The Streamlit frontend talks to the FastAPI backend only through REST calls. The backend handles inference and operational APIs, while MLflow stores experiments and model versions. Prometheus scrapes backend metrics, Grafana visualizes them, and Alertmanager pushes active alerts back into the application flow. Separately, Airflow manages the short feedback-retraining loop, while DVC manages the full reproducible training pipeline.



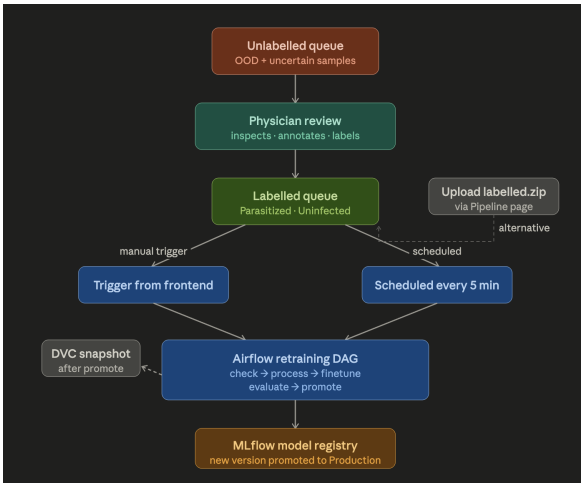
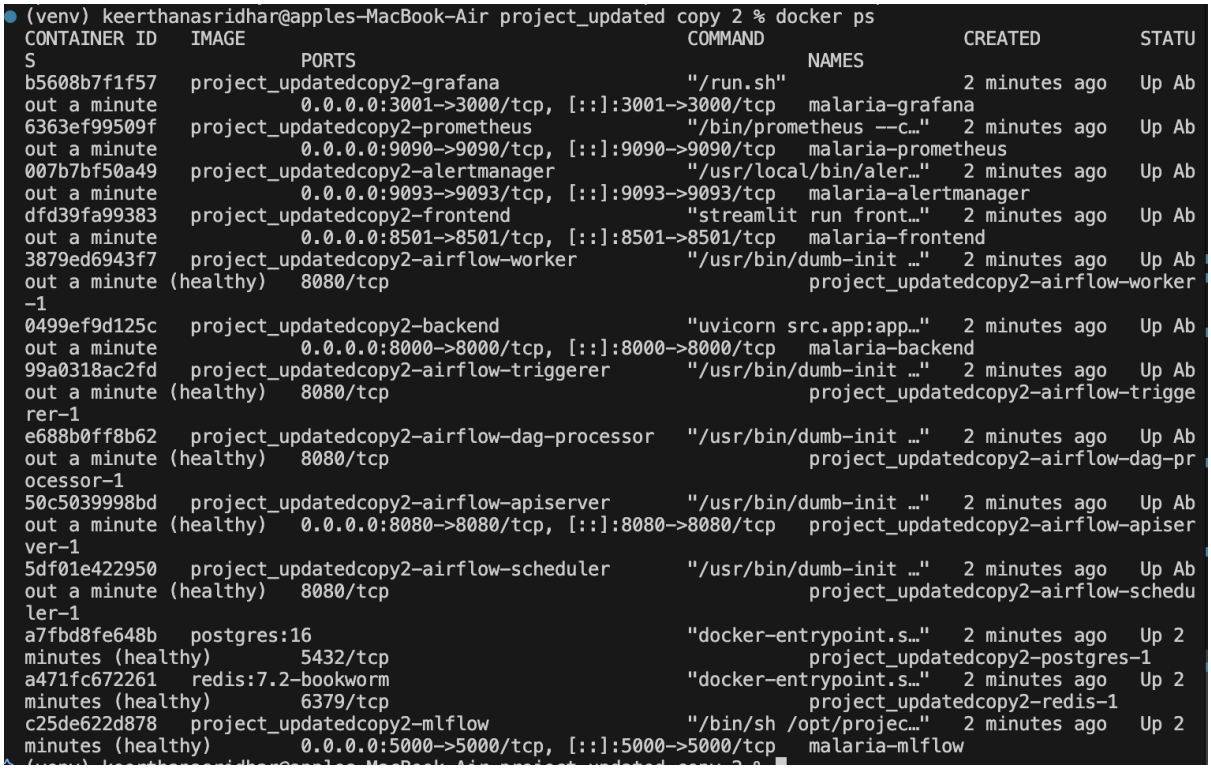


Figure 1: These two diagrams summarize the operating logic of the system. The first image focuses on the inference and alert path: a clinician uploads an image, the backend serves the MLflow Production model, Prometheus scrapes runtime metrics, Grafana visualizes them, and Alertmanager returns important alerts to the UI. The second image shows the human-in-the-loop retraining loop, where uncertain or OOD samples are reviewed by a physician, sent to the labelled queue, and then used by the Airflow DAG to produce a new promoted model version.

6.2 Containerized Deployment

The project is packaged as a multi-container system using Docker Compose. The important services include a frontend, backend, MLflow server, Prometheus, Grafana, Alertmanager, Redis, PostgreSQL for Airflow, and several Airflow components such as the scheduler, worker, API server, triggerer, and DAG processor.



```
(venv) keerthanasridhar@apples-MacBook-Air project_updated copy 2 % docker ps
CONTAINER ID   IMAGE                                          COMMAND                  CREATED         STATU
S                      PORTS                                                 NAMES
b5608b7f1f57   project_updatedcopy2-grafana                   "/run.sh"                2 minutes ago   Up Ab
out a minute           0.0.0.0:3001->3000/tcp, [::]:3001->3000/tcp   malaria-grafana
6363ef99509f   project_updatedcopy2-prometheus                "/bin/prometheus --c…"   2 minutes ago   Up Ab
out a minute           0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp   malaria-prometheus
007b7bf50a49   project_updatedcopy2-alertmanager              "/usr/local/bin/aler…"   2 minutes ago   Up Ab
out a minute           0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp   malaria-alertmanager
dfd39fa99383   project_updatedcopy2-frontend                  "streamlit run front…"   2 minutes ago   Up Ab
out a minute           0.0.0.0:8501->8501/tcp, [::]:8501->8501/tcp   malaria-frontend
3879ed6943f7   project_updatedcopy2-airflow-worker            "/usr/bin/dumb-init …"   2 minutes ago   Up Ab
out a minute (healthy)   8080/tcp                                          project_updatedcopy2-airflow-worker
-1
0499ef9d125c   project_updatedcopy2-backend                   "uvicorn src.app:app…"   2 minutes ago   Up Ab
out a minute           0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp   malaria-backend
99a0318ac2fd   project_updatedcopy2-airflow-triggerer         "/usr/bin/dumb-init …"   2 minutes ago   Up Ab
out a minute (healthy)   8080/tcp                                          project_updatedcopy2-airflow-trigge
rer-1
e688b0ff8b62   project_updatedcopy2-airflow-dag-processor     "/usr/bin/dumb-init …"   2 minutes ago   Up Ab
out a minute (healthy)   8080/tcp                                          project_updatedcopy2-airflow-dag-pr
ocessor-1
50c5039998bd   project_updatedcopy2-airflow-apiserver         "/usr/bin/dumb-init …"   2 minutes ago   Up Ab
out a minute (healthy)   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   project_updatedcopy2-airflow-apiser
ver-1
5df01e422950   project_updatedcopy2-airflow-scheduler         "/usr/bin/dumb-init …"   2 minutes ago   Up Ab
out a minute (healthy)   8080/tcp                                          project_updatedcopy2-airflow-schedu
ler-1
a7fbd8fe648b   postgres:16                                    "docker-entrypoint.s…"   2 minutes ago   Up 2
minutes (healthy)        5432/tcp                                          project_updatedcopy2-postgres-1
a471fc672261   redis:7.2-bookworm                             "docker-entrypoint.s…"   2 minutes ago   Up 2
minutes (healthy)        6379/tcp                                          project_updatedcopy2-redis-1
c25de622d878   project_updatedcopy2-mlflow                    "/bin/sh /opt/projec…"   2 minutes ago   Up 2
minutes (healthy)        0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   malaria-mlflow
```

Figure 2: The Docker process listing confirms that the system is deployed as a multi-service application rather than a single script. The screenshot shows the frontend, backend, MLflow, Prometheus, Grafana, Alertmanager, Redis, PostgreSQL, and multiple Airflow services all running together.

7 User-Facing Application

7.1 Home Page and User Experience

The Streamlit frontend is the most visible part of the system for a non-technical evaluator. The Home page is designed to be straightforward: the user sees the application title, backend connectivity state, system alert status, and an upload control for individual images or ZIP files.

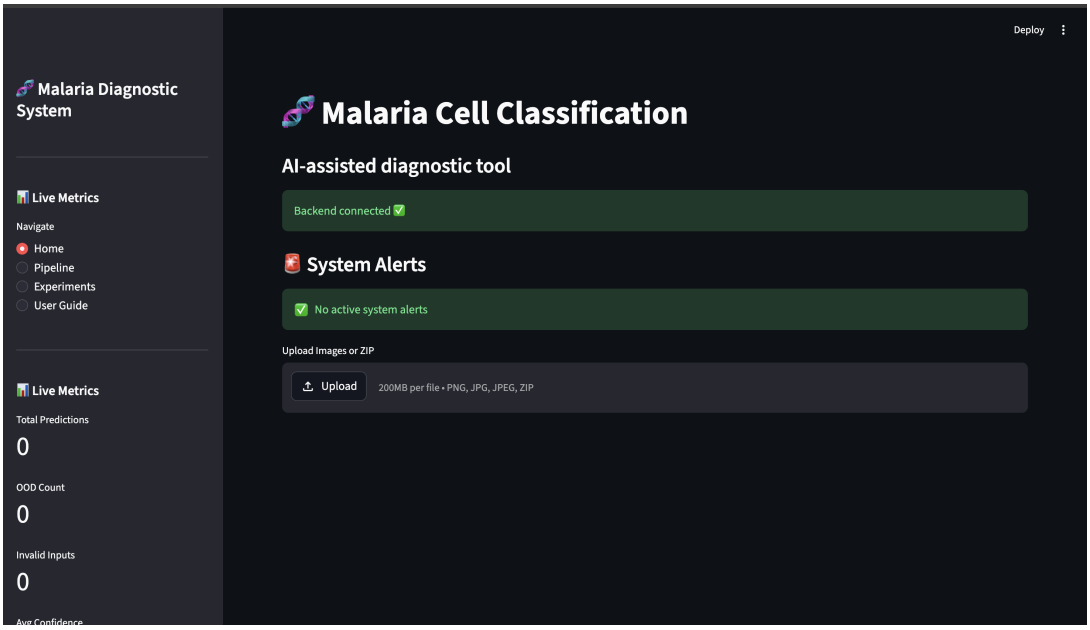


Figure 3: The Home page shows a clean starting point for a non-technical user. It makes the system state visible immediately by showing backend connectivity, active system alerts, and the file upload area.

Once an image is uploaded, the user can preview it and run analysis. The response is presented in a readable table with prediction, confidence, latency, status, and a message field.

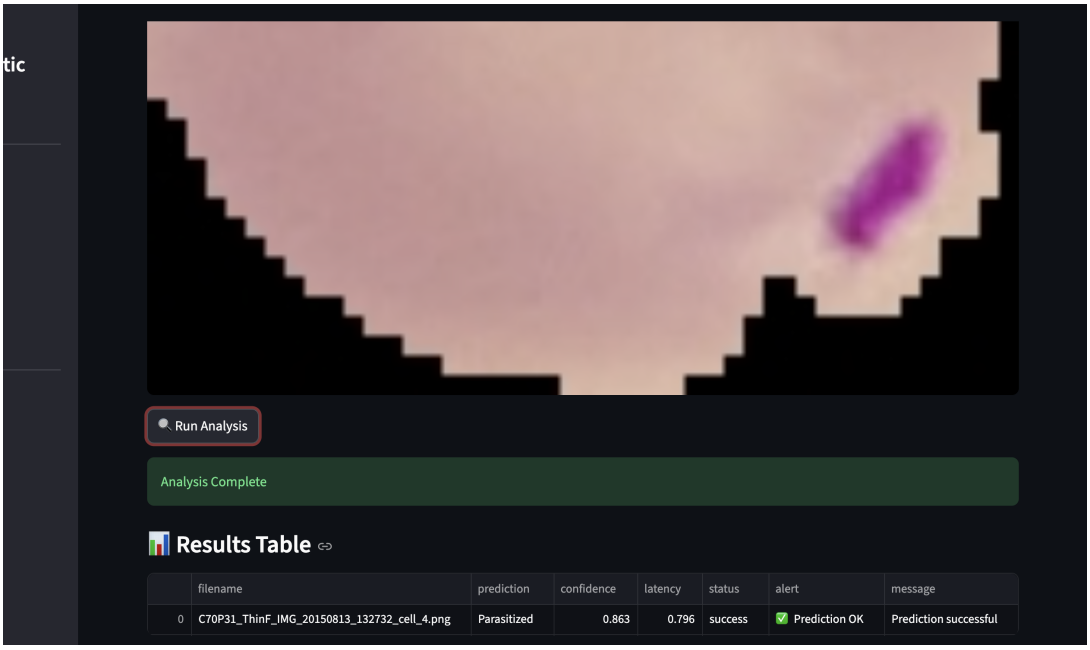


Figure 4: This screenshot shows a successful end-to-end inference from the user's perspective. The uploaded image is visible, the analysis button is clearly exposed, and the results table explains what the model predicted, how confident it was, how long inference took, and whether the overall status was successful. The table format also makes the application easier to demo because all key outputs are visible in one place.

7.2 Sidebar Metrics and Failure Visibility

The application exposes live metrics, OOD counts, invalid input counts, confidence summaries, and a failure log directly in the UI.

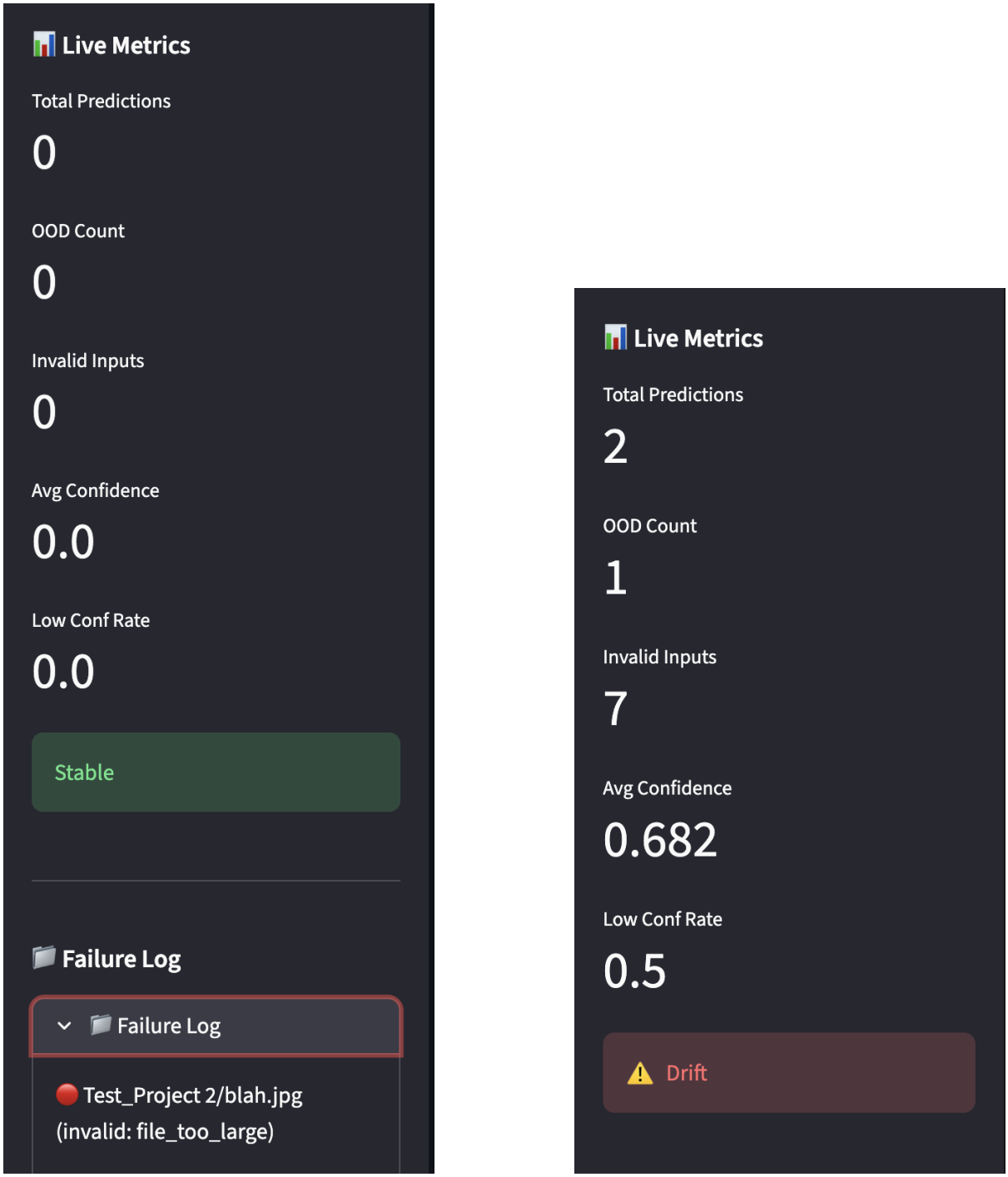


Figure 5: The left image shows the sidebar in a calm, healthy state with zero counts and a stable model status. The right image shows the same interface after real usage, where total predictions, OOD cases, invalid inputs, confidence average, and drift become visible.

7.3 Experiments and Model Metadata

The frontend also includes a dedicated Experiments page that surfaces the currently served model information. This includes the model name, run ID, run name, version, and refresh state.

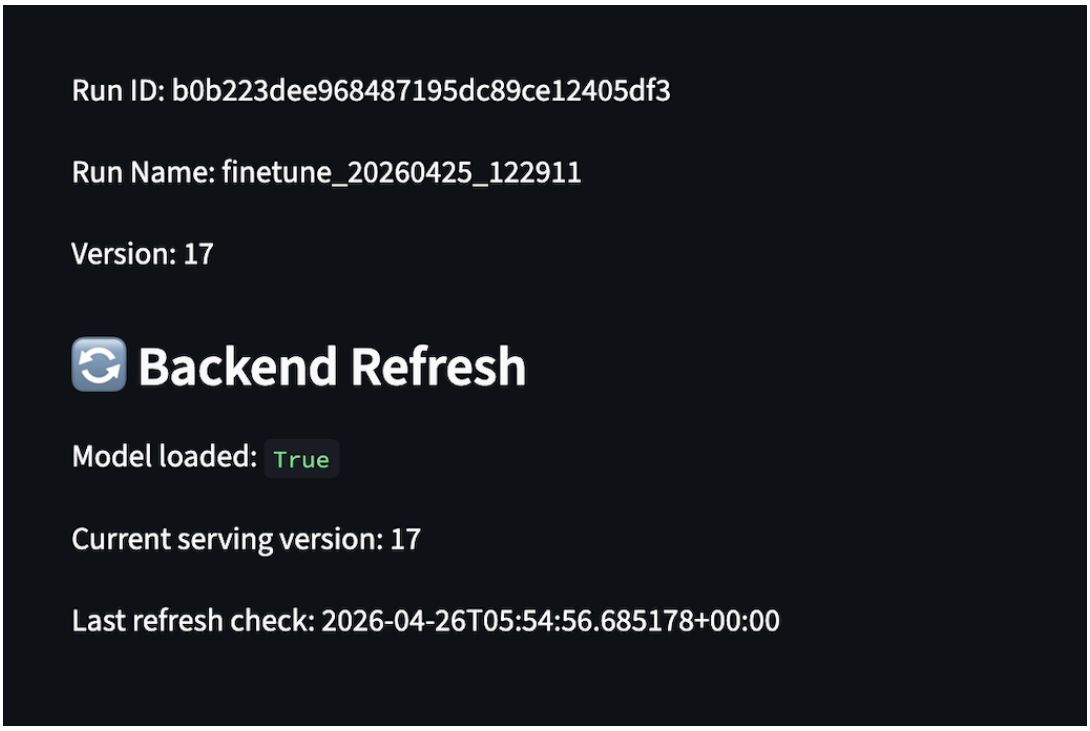


Figure 6: The Experiments page bridges the gap between the UI and the MLOps back end. It shows which model is currently being served, which run produced it, and which model version is active.

8 Data Pipeline and Reproducibility

8.1 Dataset Summary and EDA

The repository contains a clear EDA stage in `src/eda.py`. It computes class distribution, image size statistics, corrupt image counts, and channel statistics that are later reused for drift and OOD checks. Based on the current generated metrics, the dataset is balanced, with 13,816 images in each class, for a total of 27,632 images. The code also reports average width and height, as well as two corrupt images.

This stage matters for two reasons. First, it gives a quantitative picture of the dataset before training. Second, it provides baseline statistics for later runtime monitoring.

8.2 DVC-Managed Pipeline

The full reproducible workflow is defined in `dvc.yaml`. The stages are `merge_feedback`, `eda`, `preprocess`, `resize`, `train`, and `evaluate`. The preprocessing code splits the raw data into train, validation, and test sets using the configured ratios 0.70, 0.15, and 0.15. Images are then resized to 224×224.

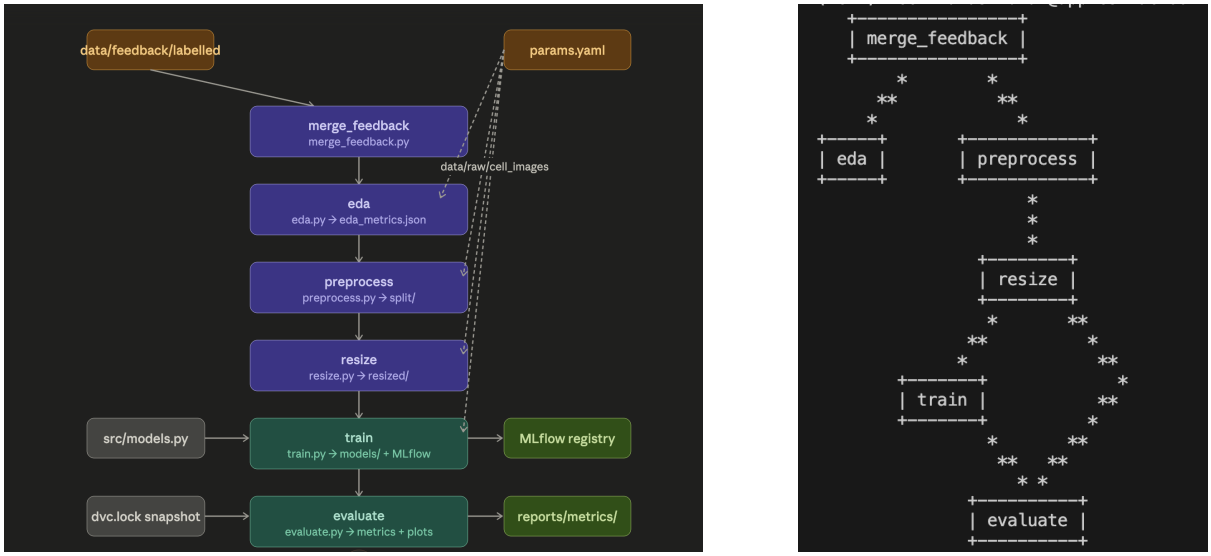


Figure 7: The block diagram on the left explains the logical flow of the reproducible ML pipeline, including feedback merging, EDA, preprocessing, resizing, training, evaluation, and MLflow registration. The DVC DAG screenshot on the right acts as proof that the stages are not only described conceptually but are actually encoded in a dependency graph that DVC can execute and track.

9 Model Development and Evaluation

9.1 Model Choice

The active model is a custom CNN defined in `src/models.py`. The configuration currently uses convolution channel sizes $[16, 32]$, kernel size 3, and a fully connected layer of size 128. A ResNet-18 option also exists in the repository, which shows that the codebase supports comparative experimentation rather than locking the system to one fixed model design.

9.2 Training and Evaluation Results

Training is implemented in `src/train.py`, and evaluation is handled in `src/evaluate.py`. MLflow logs parameters, metrics, and artifacts from both stages. The current generated metrics show solid performance:

Split	Accuracy	Precision	Recall	F1-score
Validation	0.9100	0.8804	0.9488	0.9134
Test	0.9219	0.9027	0.9457	0.9237

Table 1: Current validation and test performance from the repository outputs

These numbers show that the model is performing well on the held-out test set while also keeping a strong recall, which is especially relevant in a medical-imaging setting where missed positives can be costly.

10 MLflow Experiment Tracking and Model Registry

The project uses MLflow to log the models and their artifacts. Training runs log loss, F1, accuracy, precision, and recall. Parameters such as batch size, epochs, learning rate, and model choice are recorded. Artifacts such as model files and metrics folders are stored with each run. The model is also registered and promoted through MLflow's registry.

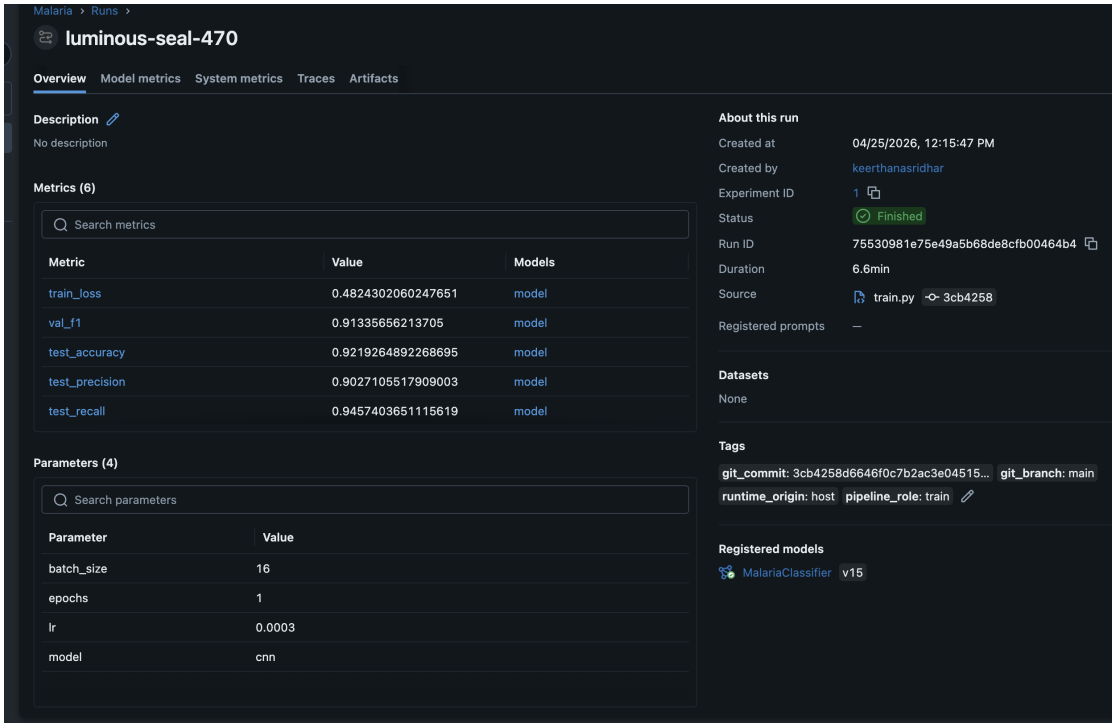


Figure 8: This MLflow run page is a strong piece of evidence because it connects code, parameters, metrics, and registry state in one place. The screenshot shows that training loss, validation F1, and test metrics were logged, while the right panel records run metadata such as the source file, Git-related tags, runtime origin, and the registered model version.

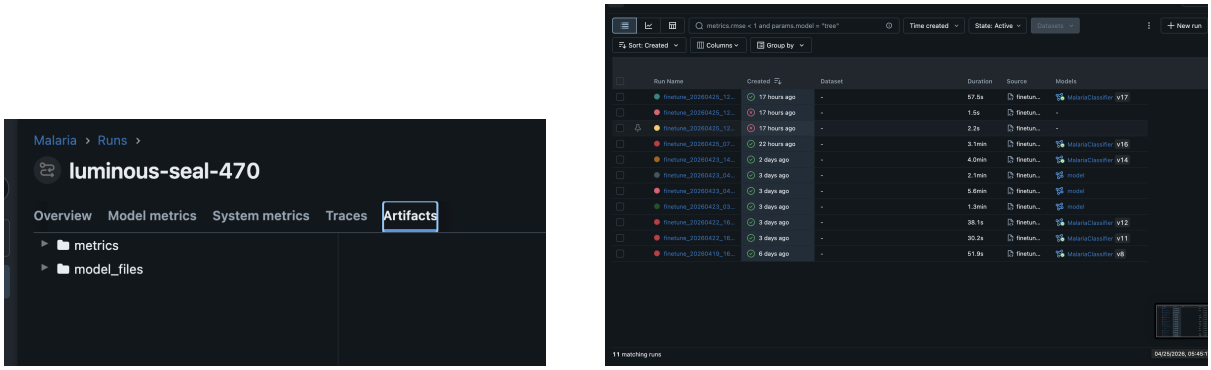


Figure 9: The left image shows the stored artifacts for a tracked run, which is important because it proves that MLflow is not only storing scalar metrics but also preserving structured outputs for later inspection. The right image shows the fine-tuning experiment history, including multiple incremental runs and promoted versions.

11 Inference, Alerts, and Monitoring

11.1 Runtime Monitoring in the UI

The frontend also surfaces alerts, aggregated confidence, total predictions, and drift signals. That makes the application more transparent and easier to evaluate in real time.

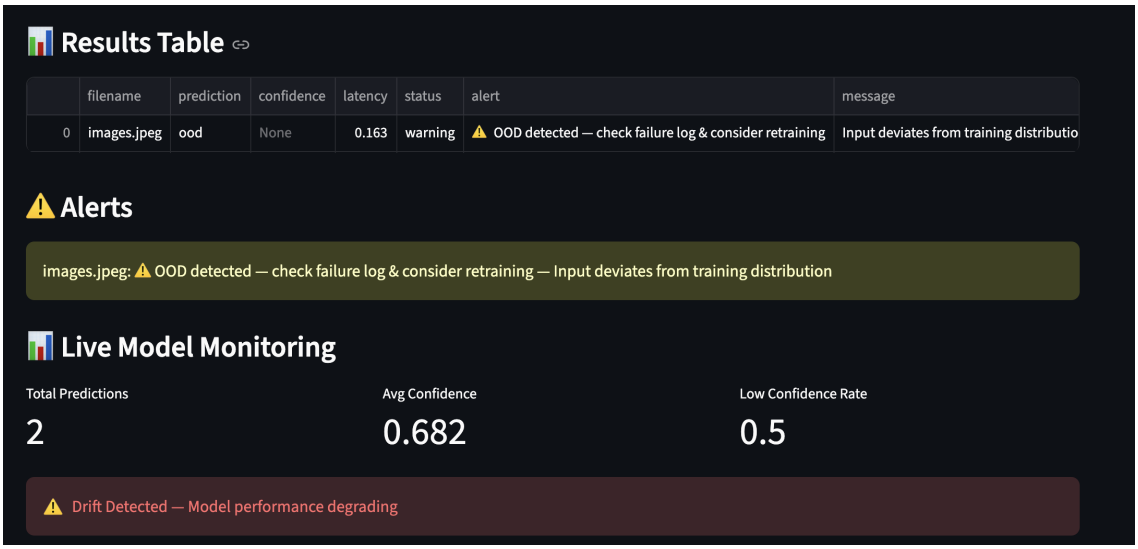


Figure 10: The results table marks the prediction as OOD, the alerts section explains that the sample deviates from the training distribution, and the live monitoring cards show the total predictions, average confidence, low-confidence rate, and drift warning.

11.2 Prometheus and Alertmanager

Prometheus scrapes backend metrics from the `/metrics` endpoint, and the alert rules watch for drift, OOD rates, high error rates, slow inference, and high invalid-input rates. When one of these rules fires, Alertmanager groups and routes the alerts.

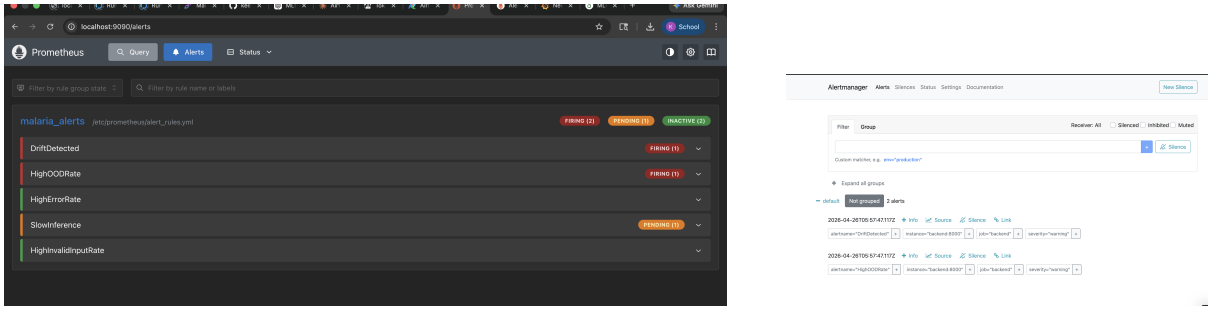


Figure 11: The Prometheus view on the left shows the alert rules in action, including DriftDetected and HighOODRate in the firing state and SlowInference in the pending state. The Alertmanager view on the right confirms that those alerts are grouped and routed downstream.

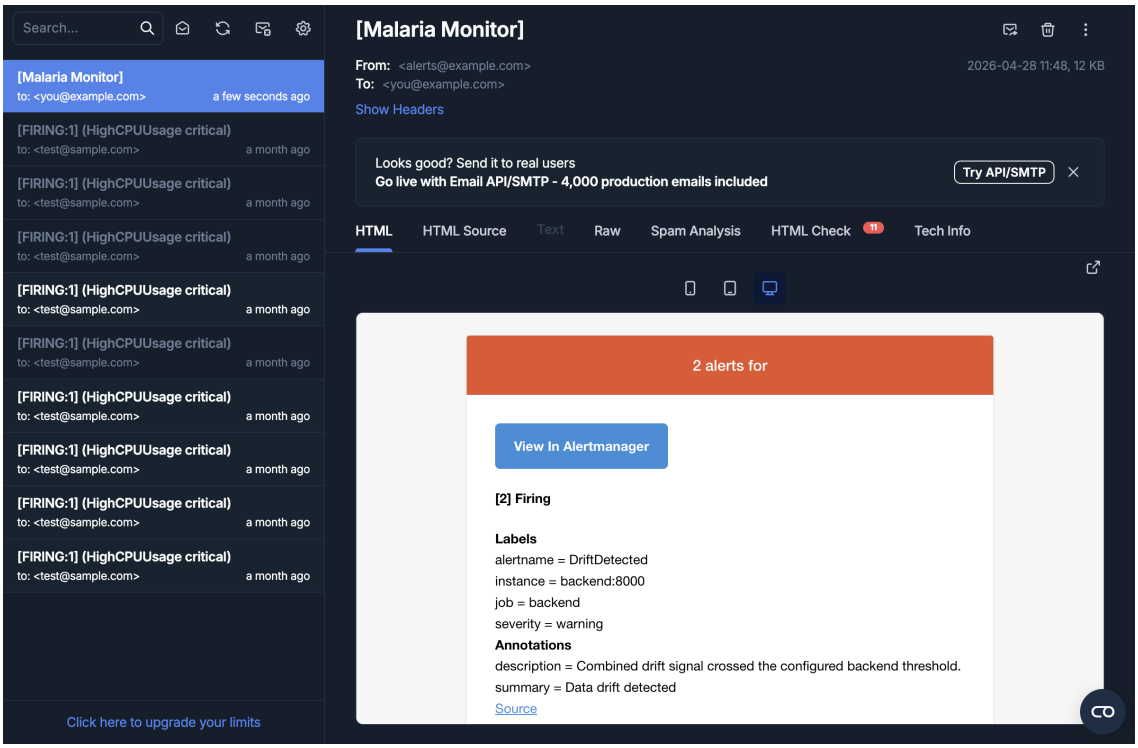


Figure 12: Mailtrap screenshot of incoming alert email.

11.3 Grafana Dashboards

Grafana adds the final layer of visibility by turning raw metric streams into understandable charts and indicators.

Figure 13: The Grafana panels show two complementary views of the backend. The left screenshot focuses on system-side and confidence-related telemetry, including CPU usage and average confidence over time. The right screenshot focuses on risk signals such as OOD rate and drift status.

12 Human-in-the-Loop Feedback and Retraining

When inference is uncertain or OOD, samples are stored in feedback queues. A physician or expert can review them, label them, and either place them in the labelled folder structure or upload them as a ZIP archive from the Pipeline page. After that, the Airflow DAG can process the new feedback and trigger fine-tuning.

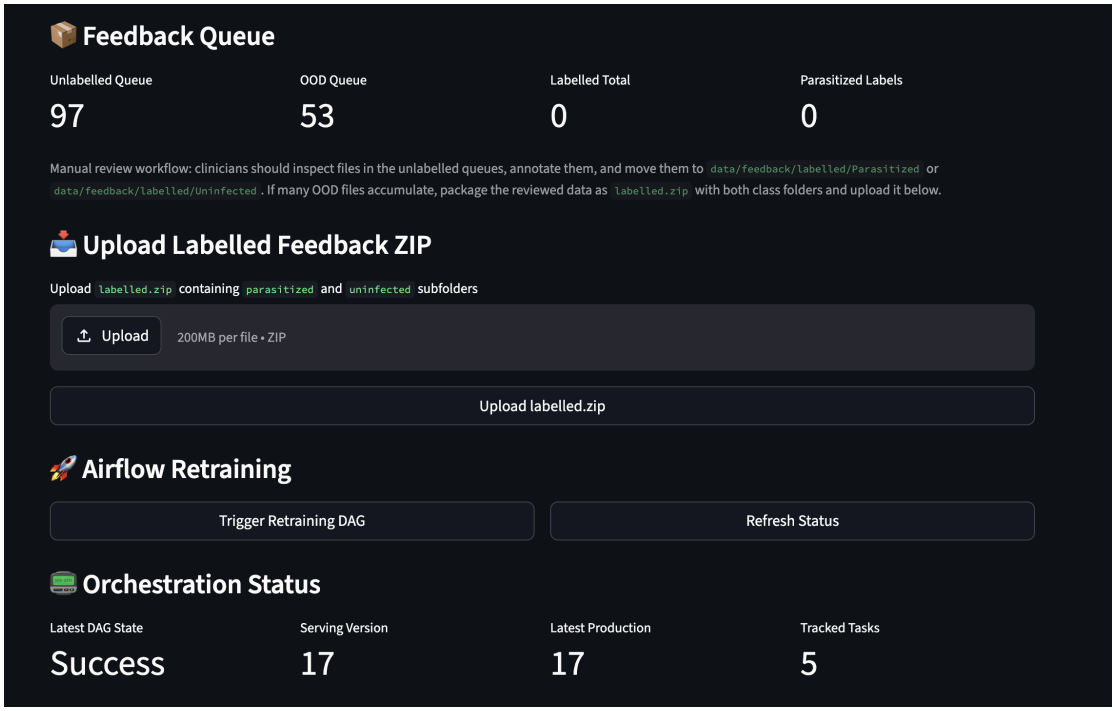
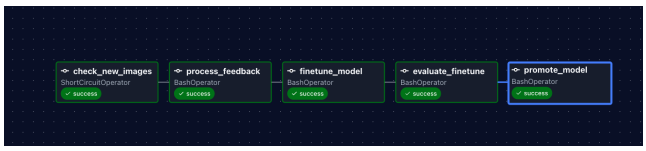


Figure 14: This screenshot shows the Pipeline page acting as an operational console rather than just a decorative dashboard. The top section exposes the current feedback queue counts, the middle section allows labelled feedback to be uploaded, and the lower section provides buttons for manual DAG triggering and status refresh. The orchestration cards at the bottom then report the latest DAG state, serving version, latest production version, and tracked task count.

The retraining pipeline itself is orchestrated through Airflow. It checks for new labelled data, processes feedback, fine-tunes the model, evaluates the candidate model, and promotes it if the validation threshold is satisfied.



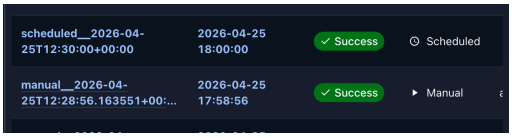


Figure 15: The Airflow screenshots show both the task structure and the operational history of the retraining DAG. The graph view on the left confirms the task sequence `check_new_images` → `process_feedback` → `finetune_model` → `evaluate_finetune` → `promote_model`. The run-history view on the right shows that both scheduled and manually triggered runs complete successfully.

The current fine-tune evaluation output reports a validation accuracy of 0.8139 against a threshold of 0.75, and the candidate was accepted. This means the feedback-driven path is not only implemented, but has also succeeded in producing a promotable model version.

13 Backend API Design

The backend in `src/app.py` exposes more than a single `/predict` route.

Endpoint	Purpose
`/health`	Liveness check for the backend service
`/ready`	Readiness check for model-serving state
`/predict`	Image inference endpoint for uploaded files
`/api/metrics`	Runtime metrics exposed to the UI and monitoring tools
`/api/failures`	Access to failure and review-log entries
`/api/feedback/stats`	Counts for unlabeled, OOD, and labelled queues
`/api/feedback/upload-labelled-zip`	Upload of physician-reviewed labelled data
`/pipeline/status`	DVC workspace state and reproducibility helper information
`/retrain`	Background trigger for the DVC reproducible workflow
`/api/retraining/trigger`	Manual trigger for the Airflow retraining DAG
`/api/retraining/status`	Latest DAG status and per-task information
`/api/model/refresh`	Manual backend refresh of served model state
`/model/info`	Metadata for the currently served model version
`/alerts`	Active alerts currently visible in the application
`/internal/alert`	Webhook receiver used by Alertmanager

Table 2: Primary backend API surface

14 Testing and Software Engineering Quality

The repository contains structured design documents in `docs/`, including architecture, high-level design, low-level design, user manual, reproducibility notes, acceptance criteria, and a test plan.

Automated testing is also present. The file `tests/test\_api.py` currently defines nine tests covering health, readiness, model info, invalid file handling, empty file handling, valid prediction, DVC status parsing, invalid-log counting, and model refresh logic.

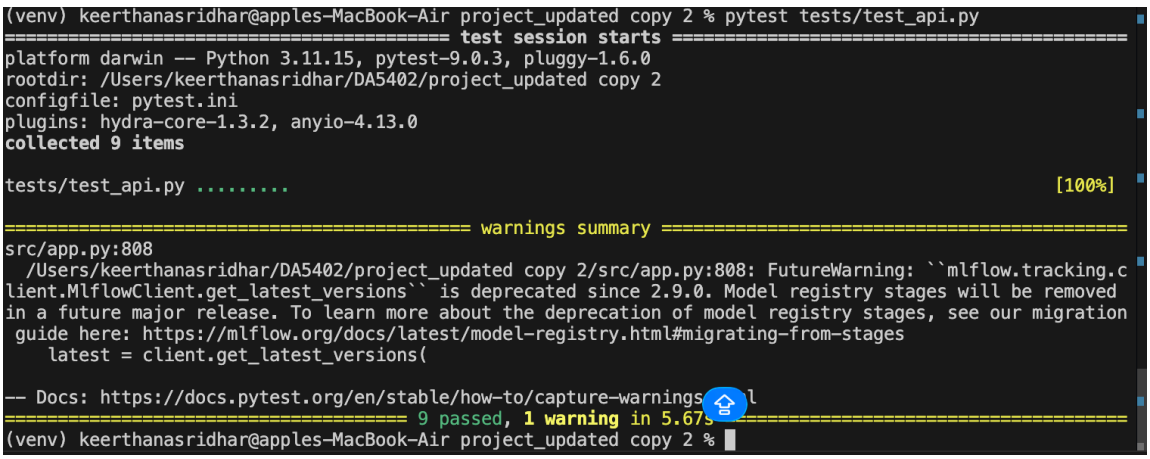


Figure 16: The pytest output shows that nine backend tests pass successfully.

A AI Disclosure

- 1) produce a boilerplate of my backend with the necessary basic endpoints.

- 2) produce a boilerplate of my frontend which is suitable for my basic backend
- 3) how do i implement airflow retriggering via a frontend button?
- 4) mlflow sever vs mlflow local, how does the mapping happen?
- 5) can i use the files absolutely saved from within a docker work on user? if not suggest a code fix for the same as im not able to access the mlflow trains within from my local computer
- 6) give me a script for the eda- involving pixel intensity and width and height
- 7) how does the decoupling of docker files for front and back end work?
- 8)i have a unzip script already. can you make it robust enough to handle both archive layouts and put everything into data/raw/cell$_images$
- 9) my app should still work even if mlflow has nothing in it. can you make sure the backend falls back to a local checkpoint cleanly 9) how can i promote my model to production via code? give me a sample
- 10) what are the good test cases for this project? give me a couple of basic test cases
- 11) generate a sample script tat lets me push my data from labelled folder to unlabelled folder
- 12) how can i generate a button, that lets the user upload a labelled.zip, that should dynamically check for (parasitized and uninfected subfolders) and extract those to data/feedback/labelled
- 13) this mlflow server should be started whenever we run the app? explain again?
- 14) give me a section in my backend that can invalidate corrupt or invalid images
- 15) how can i keep the metrics as dynamic and refreshable?
- 16) give me some guidance on how to log metrics

  The rest of the prompts used were for troubleshooting errors and bugs in the code. AI was also used to generate the readme and the report template.